Souvik Banerjee

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Experience

Amla Labs *Oakland, CA*
Founder · South Park Commons · Pre-seed Mar 2026 - Present

- Building the security layer for AI agents: authorization, isolation, and network controls for autonomous systems.
- Thesis: bring proven operating systems security models to agents — capability-based fine-grained authorization, microVM isolation, and network programmability.
- Shipped `amla-sandbox`, a 13MB WebAssembly runtime giving agents capability-scoped POSIX execution with deterministic replay; cuts LLM context overhead by ~98% versus raw tool-call patterns.

Tudor Investment Corporation *Oakland, CA*
Quantitative Developer: Macro Pipeline Nov 2024 - Jun 2026

- Quantitative research and development supporting low-latency trading strategies and systems.
- Architected and deployed a distributed system to transmit predictive signals across exchanges, credited with unlocking $10M+ in annual P&L through cross-venue strategy execution.
- Engineered a custom market data ingestion pipeline, streaming compressed packet capture files directly from AWS S3 and eliminating intermediate decompression; reduced annual cloud processing and storage costs by approximately $250,000.

Apple *Cupertino, CA*
Senior Software Engineer: Core OS Kernel Technologies Sep 2019 - Nov 2024

- Directed engineering effort to integrate ARMv9 Memory Tagging Extensions (MTE) into iOS secure kernel and firmware.
- Developed device driver infrastructure for the Apple N1 wireless chip, shipped in iPhone Air.
- Led teams building novel confidential compute operating systems, shipped in iOS 17 to >1B devices.
- Architected and delivered new features for third party device drivers on iPad, adding support for custom USB and Thunderbolt devices. Presented at WWDC 2022.
- Designed and implemented core security and stability enhancements for Apple operating systems.

Software Engineer Intern: Core Kernel May 2018 - Aug 2018

University of Texas at Austin *Austin, TX*
Undergraduate Researcher Aug 2017 - Aug 2019

- Prototyped new blockchain architectures with a focus on network and storage performance.
- Publications:
  - Ponnapalli, Soujanya, Aashaka Shah, **Souvik Banerjee**, Dahlia Malkhi, Amy Tai, Vijay Chidambaram, and Michael Wei. "RainBlock: Faster Transaction Processing in Public Blockchains." In 2021 USENIX Annual Technical Conference (USENIX ATC 21).

Teaching Assistant Aug 2016 - Dec 2017

- Taught CS 439H Operating Systems Honors, CS 429H Computer Architecture Honors, CS 311 Discrete Mathematics.

Point72: Cubist Systematic Strategies *New York, NY*
Research Analyst Intern: FX Options Trading Dec 2018

Citadel Securities *Chicago, IL*
Software Engineer Intern: Low Latency Sep 2018 - Nov 2018

Facebook *Menlo Park, CA*
Software Engineer Intern: WhatsApp iPhone May 2017 - Aug 2017
Software Engineer Intern: WhatsApp VoIP May 2016 - Aug 2016

Education

University of Texas at Austin *May 2019*
B.S. Computer Science Honors, Mathematics GPA: 3.989

- **Turing Scholar, Distinguished College Scholar, University Honors**
- Honors Thesis: Implementing Stateless Clients in Ethereum

Selected Programs

Frontier AI Security Residency *Cambridge, UK*
Research fellowship to secure frontier AI Oct 2026 - Dec 2026

M3 Fellowship *London, UK*
Fellowship for senior security professionals advancing AI security research Sep 2026 - Oct 2026

AI Security Bootcamp *Las Vegas, NV*
Intensive program for security professionals shaping how we secure emerging AI systems Aug 2026